

Tokens Per 2 MB Page for Our Models

$T_{\text{Qwen-14B}} = 819$	(MHA)
$T_{\text{Mistral-Nemo-12B}} = 4096$	(GQA)
$T_{\text{DeepSeek-V2-Lite}} = 1820$	(MLA)
$T_{\text{Mistral-Nemo-12B Synthetic MLA}} = 5461$	(Synthetic MLA)

Model	Architecture	Tokens per 2 MB page (T)
Qwen-14B	MHA	819
Mistral-Nemo-12B	GQA	4096
DeepSeek-V2-Lite	MLA	1820
Mistral-Nemo-12B Synthetic MLA	Synthetic MLA	5461

Interpretation

- Larger T means more tokens fit in each 2 MB page.
- Larger T also pushes the tail-waste curve farther to the right.
- This is why a more memory-efficient attention layout can still need a longer context before fragmentation becomes insignificant.